

LiNbOCl₄ — crystalline oxyhalide benchmark

Testing transfer of the same workflow to mixed O²⁻/Cl⁻ chemistry

1 Literature

Tanaka et al. (2023)
RT $\sigma = 10.4 \text{ mS cm}^{-1}$
Experimental $E_a = 0.240 \text{ eV}$

2 Model & software

Periodic oxyhalide model
224 atoms; 32 Li
MACE / SevenNet / M3GNet
LAMMPS; 1 GPU per run

3 MD conditions

T: 600 / 800 / 1000 / 1200 K
Ensemble: fixed-cell NVT
dt: 1 fs
Equil.: 50 ps
Production: 500 ps

4 Analysis

Li MSD $\rightarrow D(T)$
 $\ln(\sigma_{\text{NET}})$ vs $1000/T$
Arrhenius fit $\rightarrow E_a$
300 K extrapolation

5 Conclusion

Experimental slope included
Model ordering differs
from Li₃YCl₆